

ProductionManagementAI — Product master — Detailed Design Document (詳細設計書)

004_DD specifies SCR-004 under approved WI-006 plan revision 3. It implements [004_BD](#) version 2, [004_DB](#) version 2, and [WI-006 brief](#) REQ-049–REQ-055 and REQ-057. The new [BD](#) and [DB](#) addenda specify effects on SCR-001–SCR-003; their detailed screen behavior belongs to the later WI-006 DD addendum. No application code has changed in this design phase.

Document control (改版履歴)

Field	Value
Document ID / version	004_DD / 2, review
Screen / work item	SCR-004 「製品マスタ」 (Product master) / WI-006
Created / updated	Codex, 2026-09-29
Source	WI-006 brief and DEC-001–DEC-011; 004_BD v2, 004_DB v2 and their approved impact addenda

Reference documents and companions

No.	Document	Purpose
1	004_BD	Navigation, actions, visible states
2	004_DB	Schema and migration
3	004_DD-API	Endpoints, request/response and errors
4	004_DD-FN	Application services and telemetry
5	004_DD-SPD	Client request and interaction sequences

The companion files above are existing drafts awaiting their individual review turns. This main DD owns screen fields, states and client modules; it does not approve those drafts. The edit form needs an authoritative indication that an order references the product so it can lock the unit. The exact API property and conflict

response are specified when 004_DD-API is reviewed. A published mockup Artifact service is unavailable; the local static HTML previews below are the reviewable visual artifacts.

Component and file organization

Path / module	Planned purpose
src/frontend/src/features/products/	Product master pages, form, API client, validation and state components
src/frontend/src/features/production-orders/	Existing order-screen effects are specified in the later WI-006 DD addendum
src/frontend/src/features/production-orders/messages.ts	Japanese labels/messages for SCR-004 and any new SCR-001 state
src/backend/ProductionManagementAI.{Domain,Application,Infrastructure,Api}/	Product entity, service/repository, EF migration, controller

The shared header and protected route are reused. Responsive components serve PC/SP; no new component kit. The existing unsaved-change guard pattern is reused for create/edit. Backend processing is owned by 004_DD-FN; API fields by 004_DD-API.

Screen layout and mockup

The local [English-captioned HTML mockup](#) and [Japanese-captioned edition](#) show populated/empty list, create, referenced-product edit, validation, retirement confirmation, success, forbidden and phone-card states. They are static previews. The [approved BD phone wireframe](#) remains the SP layout authority; this DD changes behavior inside the same regions.

D-1 共有ヘッダー 製品マスタ

D-2 一覧モード

検索 状態 検索ボタン

製品コード 製品名 単位 状態 操作

P-1001 ブレーキキャリパー 個 使用中

P-1002 ブレーキディスクローター 枚

前へ 1 / 2 次へ

D-3 登録・編集モード

製品コード P-1001 (変更不可)

製品名 [ブレーキキャリパー]

単位 個 (変更不可)

製造指示で使用中のため単位を変更できません

図面番号 []

D-6 エラー要約・項目エラー

D-4 保存 キャンセル

D-5 使用停止の確認

製品名 キャンセル 使用停止

D-6 読み込み・成功・失敗の通知

文字とアイコンで状態を伝える

Region	Items and behavior
D-1	Shared header/nav, same as 004_BD item 1.

Region	Items and behavior
D-2	List mode: search/filter, result table/cards, paging, loading/empty/error.
D-3	Form mode: SKU, name, unit and drawing number with labels/errors; SKU always read-only in edit. If referenced by any order, unit is displayed as noneditable text with an explanatory notice.
D-4	Save/Cancel; unsaved-change guard on navigation.
D-5	Retire confirmation on list, named by product and focus-contained.
D-6	Error summary and polite success status.

Field definitions and validation

Field	Create	Edit	Client rule	Server rule / result
SKU	Required editable	Read-only, sent unchanged	Trim ends; 1–50 characters	Required; <code>lower(sku)</code> unique; a changed value on PUT is rejected.
Name	Required editable	Editable	Trim ends; 1–200 characters	Required and nonblank; Unicode text retained.
Unit	Required selection	Selection only while no order references product	One of 個, 本, 枚, 台, セット, kg, m. Referenced product shows the current unit as noneditable text with Japanese reason; preserve it in edit state.	Reject unknown/empty unit and a changed unit after any order reference (REQ-057, 004_DB).
Drawing number	Optional	Editable	Trim ends; max 100 characters; blank becomes null	Same length and normalization.
Version	Not shown	Hidden application state	Loaded with record; must travel with edit/retire	Compare against <code>xmin</code> ; stale version returns a conflict.

An authoritative reference-lock indication is loaded with edit detail; the UI never infers it from retirement state or cached order counts. A retired product with no orders may still change unit; a referenced product may still change name and drawing number. A new order may reference the product after the form loads, so the server rechecks inside the edit transaction and rejects a changed unit. The client retains the draft and offers a reload. Retirement does not unlock the unit.

Validation is shown after blur or attempted submit, not on initial render or each keystroke. Required fields have visible labels; error text is linked to the control with `aria-describedby`, synchronized with `aria-invalid`, announced and summarized at submit. Invalid submit retains values and focuses the summary or first invalid field. Save remains available before validation and is disabled only while a valid request is in flight. Server validation is authoritative. Use semantic `<form>`, `<label>`, `<select>`, named controls and submit buttons; display a locked unit as read-only text with a reason rather than a disabled selector whose value could be omitted.

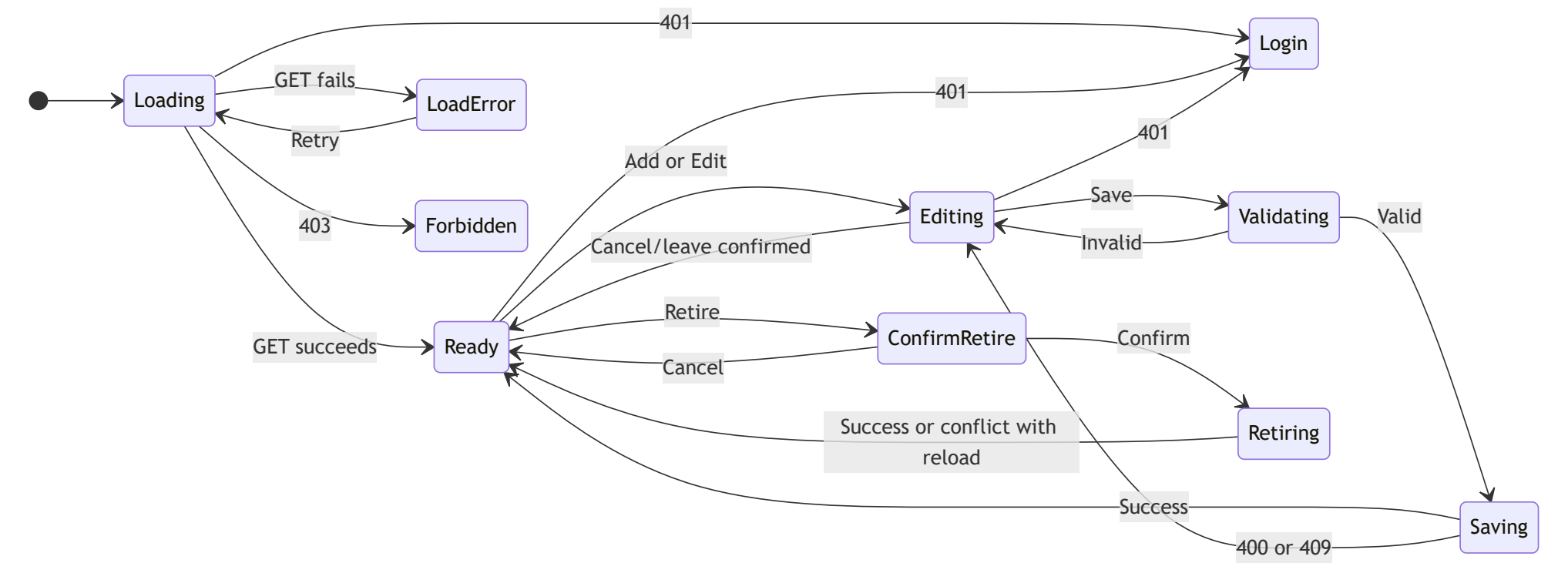
Mobile controls keep comfortable touch targets. This follows the `modern-web-guidance` forms, required-field-feedback and accessible-error-announcement guides retrieved 2026-09-29.

Loading, empty, error and success states

State	Trigger	Visible result
Loading	List/detail GET pending	Heading and loading status; dependent controls wait.
Empty catalog	Successful unfiltered list with total 0	Explain no products exist; Add stays available.
No match / page beyond end	Successful filtered or out-of-range page with no rows	Distinguish from empty catalog; offer Clear or valid page navigation.
Ready	Successful list/detail	Rows/cards, count, status text and actions.
Locked edit	Detail reports an order reference	Noneditable unit with reason; name/drawing number editable.
Fetch failure	GET network/server error	Error and 「再試行」 (Retry); do not present stale data as current.
Validation/duplicate	Invalid input or server field conflict	Retain draft; inline errors and summary; enable correction.
Stale write / newly locked unit	Version or reference conflict	Explain change and offer reload; no silent retry.
Save success	Create/edit succeeds	Return to saved list URL and announce success politely.
Retire success	Confirmation succeeds	Stay on list; show row retired and announce success politely.
Unauthorized	401 or 403	Redirect to <code>/login</code> on 401; forbidden state on 403.

If a saved product is excluded by preserved filters, the list shows a success notice and a Clear option without claiming the row is visible. If the return page is now beyond the last page, navigate to the last valid page while retaining filters.

Screen state machine



From → to	Trigger	Screen behavior
Loading → Ready / LoadError / Forbidden / Login	List/detail request	Render data, retry, permission message, or redirect, respectively.
Ready → Editing	Add/Edit	Navigate to form mode; preserve list return URL.
Editing → Validating → Editing	Invalid submission	Keep input, show error summary, focus first invalid field.
Editing → Saving → Ready	Valid save	Prevent duplicate submit, then return to list with success notice.
Saving → Editing	Field/stale/unit-lock conflict	Map field errors or show a changed-record notice and reload option.
Ready → ConfirmRetire → Retiring → Ready	Retire and confirm	Focus into dialog; on success update list; on conflict reload current row, no silent retry.
Editing → Ready	Cancel or confirmed navigation	Discard draft only after unsaved-change confirmation.

Filter/page changes and input edits keep the same route-level state and are specified in 004_DD-SPD. The diagram shows distinct route-level and mutation states; transitions that remain in the same state are table-only. Retire Cancel and Escape restore focus to the invoking button; a successful retirement moves focus to the status or updated row. The dialog names the product and contains focus while open.

Screen-owned modules

Module	Responsibility	Dependencies
ProductMasterPage	Parse URL filters, request/manage list, render table/cards and states	Product API client, shared header
ProductFormPage	Create/edit route, field values/errors, submit and dirty guard	Product API client, message catalog
RetireProductDialog	Confirm active product retirement, focus/escape behavior	Product API client, accessible dialog pattern
ProductStateLabel	Render active/retired text plus non-color cue	Japanese catalog
Existing order picker/filter	Affected-screen behavior is specified in the later WI-006 DD addendum	Extended GET /api/products rows

Accessibility, security and observability

- WCAG 2.2 AA: logical heading/focus order, visible focus, keyboard actions, text status alongside color, Japanese error summary and field messages, dialog focus containment/return, PC/SP layouts and 200% zoom. Verify with automated axe and manual keyboard/zoom review. A newly discovered unit-lock conflict is announced with a reload choice.
- API role checks remain authoritative. The UI does not infer authorization from hidden buttons. Product fields are business reference data, with no expected PII or secrets. Same-origin cookie and JSON-only mutation posture is documented in 004_DD-API.
- No SKU/name/drawing number or full request body is logged or used as a telemetry label. Outcome counts and error details are in 004_DD-FN.

API and persistence mapping

Screen action	API operation	Persistence / concurrency	Contract owner
Search and page	GET /api/product-master	Bounded read of products ; no write	004_DD-API API-PM-01
Load edit form	GET /api/product-master/{id}	Read product, version and authoritative order-reference lock indication	004_DD-API API-PM-02

Screen action	API operation	Persistence / concurrency	Contract owner
Create	POST /api/product-master	Insert active product; DB SKU uniqueness and allowed-unit constraints are final authority	004_DD-API API-PM-03; 004_DD-FN FN-028
Edit	PUT /api/product-master/{id}	Versioned product update; product-row lock and order-reference check in one transaction	004_DD-API API-PM-04; 004_DD-FN FN-029
Retire	POST /api/product-master/{id}/retire	Versioned <code>is_active=false</code> ; no delete or automatic reactivation	004_DD-API API-PM-05; 004_DD-FN FN-030

The later API companion owns exact JSON names, Problem Details error codes and request timeouts; the FN companion owns SQL/EF lock sequence and OpenTelemetry span/metric names. Both must agree with the approved 004_DB transaction rule before dependent code. Do not transmit or log product names, SKUs, drawing numbers, full bodies, cookies or version tokens in telemetry labels. Same-origin cookie authentication and JSON-only mutations follow the existing security posture.

Test viewpoints

See [WI-006 test plan](#) for TC-303–TC-307, TC-310–TC-312 and TC-314: catalog list/search, create validation and duplicates, edit/stale version, retirement, role denial, seed migration, responsive and accessibility behavior. REQ-057 and affected-screen requirements need additional scenarios when the test plan is updated after the last design review. No application test has run. The precise reference-lock DTO field and conflict mapping are an outstanding *technical contract* for 004_DD-API, not an unresolved business decision.